晶体点群

一、点群操作

至少有一个点保持不动 for any operation

{ objects move
  basis still } active

{ objects still
  basis move } passive

References:

space groups for solid state scientists, (2013) M. Glazer G. Burns

Group theory: Application to the physics of condensed matter

Mildred S. Dresselhaus

Gene Dresselhaus

Ado Jorio

Springer

active perspective :

$\vec{r}' = R\vec{r}$

1° E $\quad n=1$

2° proper rotation

n	$\frac{2\pi}{n}$
$n=2$	π
$n=3$	$2\pi/3$
$n=4$	$\pi/2$
$n=6$	$\pi/3$

正当转动：转动矩阵 $\det = 1$

mirror & inversion

$\det = -1$ in 3d

与所有正当转动对角

(Hermann-Mauguin)

2 3 4 6 international symbol

C_2 C_3 C_4 C_6 schoenflies symbol

③反旋操作③次 $C_3^{(3)(3)} = C_1$

转动群

主轴：n 值最大的转轴

C_2 [001]
↓ 旋转轴

2 (C_2)

3 (C_3)

4 (C_4)

6 (C_6)

3° 空间反演

$\vec{r} \to -\vec{r}$ $\quad R = \begin{pmatrix} \bar{1} & & \\ & \bar{1} & \\ & & \bar{1} \end{pmatrix}$

$\bar{1}$ (i) → C_i
inter schien

4° 镜面反射

m (σ)

先确定主轴，与主轴垂直的面 水平镜面反射 ↑ σ_h

与主轴平行的面 σ_v：主垂镜面反射

中分 XOY plane σ_d：

(对角面反射)
↓ 特殊的垂直镜面

m σ_h

$\frac{2}{m}$ σ_v

mmm （三个垂直的面反射）

5^0 那正当旋动 （旋动 + 反演）

① 间隙方法

正当旋动 n，后作空间反演

记作 $\bar{n}$ （iC_n）

希腊语词根：坚固的，立体的

stereographic projections

o：向上下

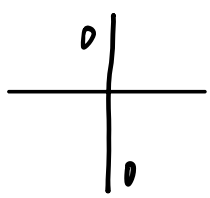
+ x：向上上

极射投影图

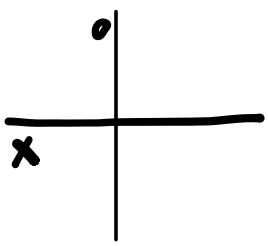


$\bar{4}$

$\bar{4}^2 = 2$

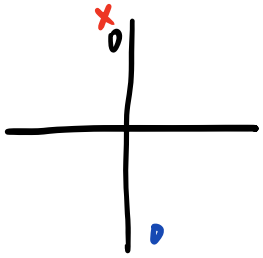


$\bar{4}^3 =$



$\bar{4}^4 = 1$

$\bar{2}$

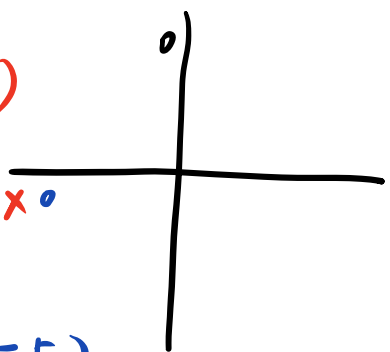


$\therefore \quad \bar{2} = m$

② Schoenflies

先施行已有旋动 C_n，后对垂直于旋动轴的面做镜向反射 σ

记作 $S_n = \sigma_h C_n$



$S_\psi \quad (\bar{\psi}^3)$

$S_\psi^3 \ (\psi)$

$S_3 \ (\bar{6}^5) \qquad S_6 \ (\bar{3}^5)$

$S_3^2 = C_3^2$

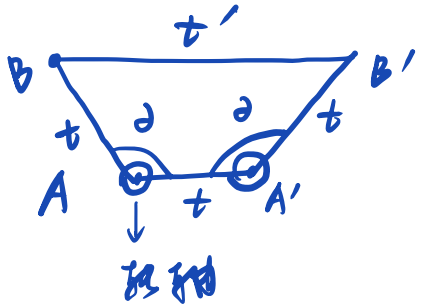
$S_3^3 = S_3 \ S_3 \ S_3 = \sigma_h$

$S_3^5 \quad (\bar{6})$

$S_6^5 \quad (\bar{3})$

3度旋转与1度旋转的晶系一样

二. 晶系 (晶系是点群概念，晶格系统是空间群概念)



A ⊙

A'移动a角到B点，说明B点是原来的晶格点

A' ⊙ 反向移动a，A → B'

B'也是从前晶格而来

$t' = mt$, $m \in \mathbb{Z}$ (晶格要求)

$2t\sin(a-\frac{\pi}{2}) + t = mt$

$\cos a = \frac{1-m}{2}$

$|\cos a| \le 1$

$m = -1, 0, 1, 2, 3$

$n = 2, 3, 4, 6, 1$

$N = 1, 6, 4, 3, 2$

只有三之外的4种选择

据此，可以分晶系。

① 三斜晶系 Triclinic

晶胞只包含 1(I) 和/或 1(i)

$\vec{r}' = \{1|\} \vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$

$\vec{r}' = \{\bar{1}|\} \vec{r} = -x\vec{a} - y\vec{b} - z\vec{c}$

对晶胞的点阵关系 不加任何限制

($\vec{a}, \vec{b}, \vec{c}$ 大小关系，角度关系)

② 单斜晶系 monoclinic

按① 增加了 2 C(C_2)，镜面反射 m(σ)

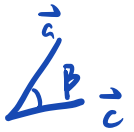
$\vec{r}' = \{2[001]|\} \vec{r} = -x\vec{a} - y\vec{b} + z\vec{c}$

$\vec{r}' = \{m[001]|\} \vec{r} = x\vec{a} + y\vec{b} - z\vec{c}$

变换前：$x\vec{a}, z\vec{c}$ $\xrightarrow[m(\sigma)]{\text{变换后}}$ $x'\vec{a}, z'\vec{c}$

垂直关系不变

$= -X \cdot z\ (\vec{x} \cdot \vec{c})$



$\cos\beta = -\cos\beta \Rightarrow \beta = \frac{\pi}{2}$

同理 $\alpha = \frac{\pi}{2}$

$\angle \vec{x}O\vec{c} = \beta$

$\angle \vec{y}O\vec{c} = \alpha$

$\angle \vec{x}O\vec{y} = \gamma$

这对 σ 也成立

如果是 $\{0\}[1]$ 或与之垂直的

镜面，$\alpha = \gamma = \frac{\pi}{2}$

③ 正交关系（orthorhombic）

按①②，增加二个二度轴

分别沿 $\vec{x}$（$[100]$）和 $\vec{y}$（$[010]$）方向

$\{2|100\}\vec{r} = x\vec{a} - y\vec{b} - z\vec{c}$

$\{2|010\}\vec{r} = -x\vec{a} + y\vec{b} - z\vec{c}$

联合操作

$\{2|100\}\{2|010\}\vec{r} = -x\vec{a} - y\vec{b} + z\vec{c}$

$= \{2|001\}\vec{r}$

两个二度轴 $\rightarrow$ 第三个二度轴

$\Rightarrow \quad \alpha = \beta = \gamma = \frac{\pi}{2}$

基矢并矢任意

④ 四方晶系 Tetragonal

增加四度旋动

C_4 或 $\bar{4}$ (S_4^3)

$$\vec{r}' = \{4\,[001]\}\,\vec{r}$$

$$= -y\vec{a} + x\vec{b} + z\vec{c}$$

$$|\vec{a}| = |\vec{b}|$$

$$\vec{r}' = \{4^3\,[001]\}\,\vec{r} = y\vec{a} - x\vec{b} + z\vec{c}$$

$$\alpha = \beta = \gamma = \frac{\pi}{2}$$

⑤ 立方晶系 Cubic

增加 4 个三度旋动，沿体对角线方向

3[111]　　以 矢量(111)方向

为轴的3度旋转

3[11$\bar{1}$]　　以矢量(11-1)方向

为轴的3度旋转

单位矢量 $u_1=\frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}$ 和 $u_2=\frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\-1\end{pmatrix}$

Euler

Rodrigues' Rotation formula

$$R(\vec{n},\theta)=\cos\theta I+(I-\cos\theta)\,nn^T+\sin\theta\,U$$

$$U=\begin{pmatrix}0&-n_z&n_y\\n_z&0&-n_x\\-n_y&n_x&0\end{pmatrix}\qquad U\vec{v}=\vec{n}\times\vec{v}$$

$$R=e^{\theta U}=I+\sin\theta\,U+(1-\cos\theta)U^2$$

$\theta=120^\circ$

$$R(\vec{u}_1,\theta)=\begin{pmatrix}0&0&0\\1&0&0\\0&1&0\end{pmatrix}\qquad R(\vec{u}_2,\theta)=\begin{pmatrix}0&1&0\\0&0&-1\\-1&0&0\end{pmatrix}$$

$\{3[111]\}\vec{r} = z\vec{a} + x\vec{b} + y\vec{c}$

$\{3^2[111]\}\vec{r} = y\vec{a} + z\vec{b} + x\vec{c}$

$\{3[11\bar{1}]\}\vec{r} = y\vec{a} - z\vec{b} - x\vec{c}$

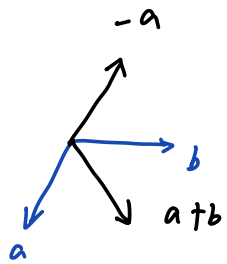
$\{3^2[11\bar{1}]\}\vec{r} = -z\vec{a} + x\vec{b} - y\vec{c}$

$|\vec{a}| = |\vec{b}| = |\vec{c}|$

$\alpha = \beta = \gamma = \frac{\pi}{2}$

⑥ 三角和六角晶系

Hexagonal 六角晶系 $6(C_6)$ 或 $\bar{6}(S_3^5)$



$\vec{r}' = \{6[001]\}\vec{r}$

$= (x-y)\vec{a} + x\vec{b} + z\vec{c}$

$= x(\vec{a}+\vec{b}) - y\vec{a} + z\vec{c}$

$\vec{r}'' = \{6^2[001]\}\vec{r}$

$= x\vec{b} + y(-\vec{a}-\vec{b}) + z\vec{c}$

$x\vec{a}\cdot y\vec{b} = x(\vec{a}+\vec{b})\cdot(-y\vec{a})$

$\Rightarrow xy|\vec{a}||\vec{b}|\cos\gamma = xy[-|\vec{a}|^2 - |\vec{a}||\vec{b}|\cos\gamma]$

$|\vec{a}| = |\vec{b}|$ $\cos\gamma = -\frac{1}{2}$

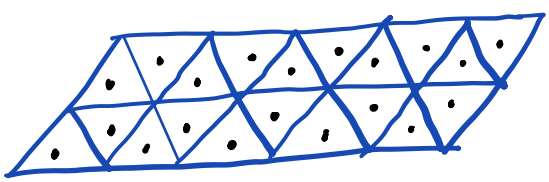
$\alpha = \beta = \frac{\pi}{2}$ $\gamma = 120°$

三角晶系 Trigonal 3 (C_3) 或 $\bar{3}$ (S_6^5)

$a = b$, $\alpha = \beta = 90°$, $\gamma = 120°$

有心的三角晶系 rhombohedral 特点的三角晶系 (菱方晶系) (不具有 b 度的动对称.)

沿 [111] 方向 (体对角线) 的唯一一个 C_3 旋转轴



· 在平面之外上下投影

$|\vec{a}| = |\vec{b}| = |\vec{c}|$

$\alpha = \beta = \gamma < 120°$

$\neq 90°$

See

space groups for solid state scientists, (2013) M. Glazer G. Burns

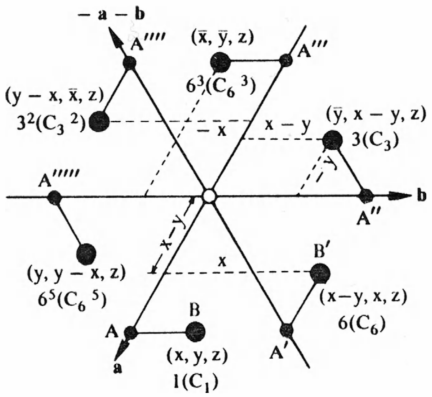


Figure 1.13 General equivalent positions with respect to hexagonal axes looking down the **c**-axis.

P21

M. Glazer

G. Burns

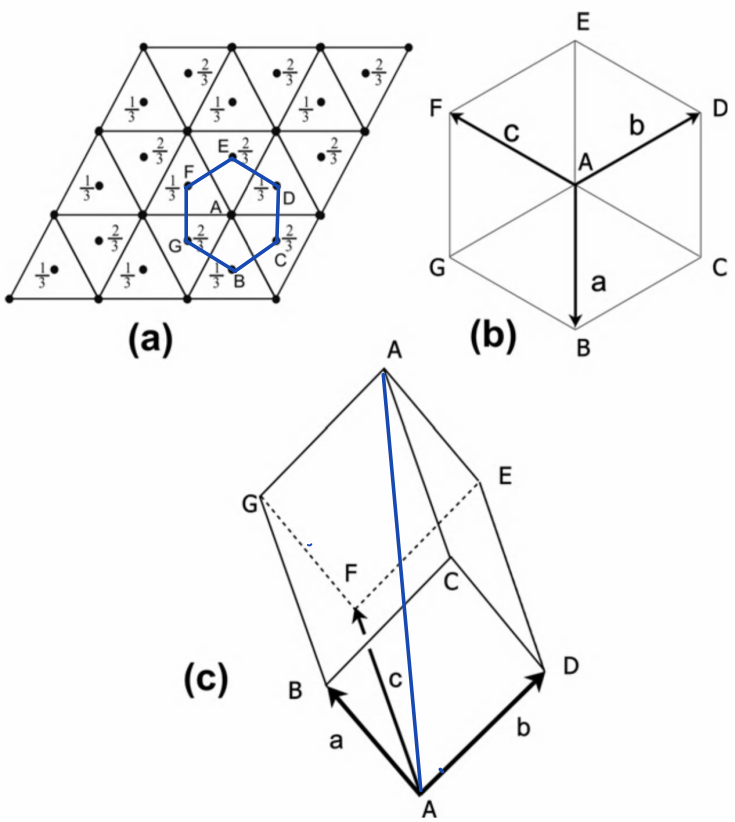


Figure 2.10 (a) The projection, down the **c**-axis, of a hexagonal lattice with additional points (centering) at (2/3, 1/3, 1/3) and (1/3, 2/3, 2/3). These points are labelled B and C, respectively, in the unit cell at the bottom right. (b) The rhombohedral unit cell looking down the hexagonal axis. (c) The rhombohedral unit cell in a 3-dimensional view. The capital letters A to G are used to indicate the relationship of the points in all three figures.

极射投影图（stereographic projections）

- 外圆实线：有水平镜面反射
- 外圆虚线：没有水平镜面反射
- 正n边形： 主轴为n度主轴
- 实心：proper rotation (行列式为1)
- 空心：improper rotation（行列式为-1）
- 圈内虚线：与主轴垂直的二度轴
- 圈内实线：垂直镜面
- 圈上小椭圆：二度轴
- ○: 面之下； ＋或×： 面之上

Stereo：希腊词根，坚固的，立体的。

画图规则约定

三. 32 种晶体点群

stereographic projections
极射投影图

1. C_n 群

(5个群)



C_1 : E



E
C_2

中心图案：2轴
= 2重轴

$\vec{r}' = \{2[001]\}\vec{r} = -x\vec{a} - y\vec{b} + z\vec{c}$

$C_{2z}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -x \\ -y \\ z \end{pmatrix}$

↓ 简记为

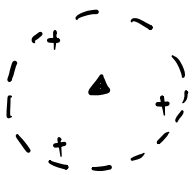
$C_{2z}(\bar{x}, \bar{y}, z)$

$E(x, y, z)$

外围实线(虚线)表示有(没有)水平镜面反射。

+ x : 面之上

o : 面之下



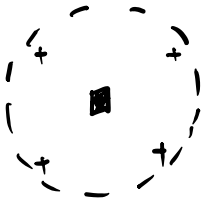
C_3 : C_3, C_3^2, $C_3^3 = E$

$C_{3z}\left(-\frac{x}{2} - \frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x - \frac{1}{2}y, z\right)$

$\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)(x + iy)$

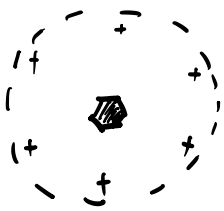
$\left(-\frac{1}{2}x - \frac{\sqrt{3}}{2}y\right) + \left(\frac{\sqrt{3}}{2}x - \frac{1}{2}y\right)i$

$C_{3z}^2\left(-\frac{x}{2} + \frac{\sqrt{3}}{2}y, -\frac{\sqrt{3}}{2}x - \frac{y}{2}, z\right) = C_{3z}^{-1}$



C_4 : C_4, $C_4^2 = C_2$, C_4^3, $C_4^4 = E$

$C_{4z}(\bar{y}, x, z)$ $C_{4z}^3(y, \bar{x}, z) = C_{4z}^{-1}$



C_6

$C_{6z}\left(\frac{x}{2} - \frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x + \frac{y}{2}, z\right)$

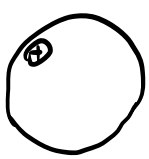
$C_{6z}^5\left(\frac{x}{2} + \frac{\sqrt{3}}{2}y, -\frac{\sqrt{3}}{2}x + \frac{y}{2}, z\right) = C_{6z}^{-1}$

C_1, C_2, C_3, C_4, C_6 5个正常旋动群（循环群）

2. C_{nh} 群（5个群）

$C_n + \sigma_h$

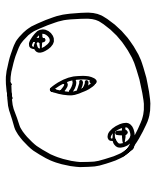
（水平镜面）



C_{1h} : E, σ_h

$\sigma_h = \sigma_{xy}(x, y, \bar{z})$

$C_{1h} = C_{1v}$ （C_1 没有主轴）

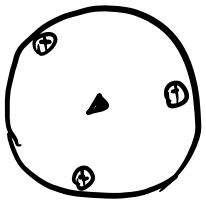


C_{2h} : C_2, σ_h, $C_2\sigma_h = i$,

$C_2^2 = \sigma_h^2 = E$

$C_{2z}\sigma_{xy}(\bar{x}, \bar{y}, \bar{z}) = i$

4个群元



C_{3h} : C_3, C_3^2, σ_h, $C_3\sigma_h = S_3$

$C_3^2\sigma_h = S_3^{-1}$, E

$C_{3z}\,\sigma_{xy}\left(-\frac{x}{2}-\frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x-\frac{y}{2}, \bar{z}\right) = S_{3z}$

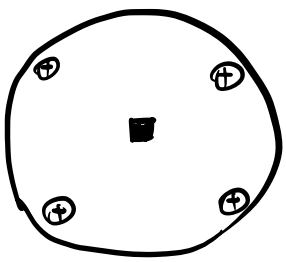
$C_{3z}^2\,\sigma_{xy}\left(-\frac{x}{2}+\frac{\sqrt{3}}{2}y, -\frac{\sqrt{3}}{2}x-\frac{y}{2}, \bar{z}\right) = S_{3z}^{-1}$

6个群元

Q: $C_3^2\sigma_h \overset{?}{=} S_3^2$

两个非正当形动变
正当形动

$S_n = \sigma_h C_n$



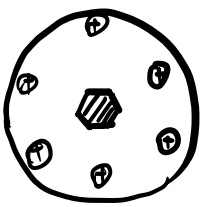
C_{4h} : C_4, C_2, C_4^3, σ_h, $C_4\sigma_h = S_4$

$C_2\sigma_h = S_2 = i$, $C_4^3\sigma_h = S_4^{-1}$, E

$C_{4z}\,\sigma_{xy}\left(\bar{y}, x, \bar{z}\right) = S_{4z}$

8个群元

$C_{4z}^3\,\sigma_{xy}\left(y, \bar{x}, \bar{z}\right) = S_{4z}^{-1}$



C_{6h} : C_6, $C_6^2 = C_3$, $C_6^3 = C_2$, $C_6^4 = C_3^2$,

C_6^5, E

12个群元

$C_6\sigma_h = S_6$, $C_3\sigma_h = S_3$, $C_2\sigma_h = i$

$C_3^2\sigma_h = S_3^{-1}$, $C_6^5\sigma_h = S_6^{-1}$, σ_h

of (o and +)

= # of group elements

从一图上找出群元

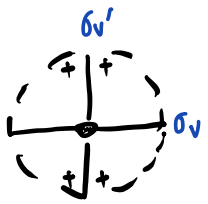
$C_{62}\ \sigma_{xy}\ \left(\frac{x}{2}-\frac{\sqrt{3}}{2}y,\ \frac{\sqrt{3}}{2}x+\frac{y}{2},\ \bar{z}\right)=s_{62}$

$C_{62}^{5}\ \sigma_{xy}\ \left(\frac{x}{2}+\frac{\sqrt{3}}{2}y,\ -\frac{\sqrt{3}}{2}x+\frac{y}{2},\ z\right)=s_{62}^{-1}$

3. C_{nv} 群　（C_n + 垂直镜面反射）

（4个群）　含有 n 度主轴和过主轴的 n 个垂直镜面 → 正和群动

$C_{1v}=C_{1h}$（已出现）



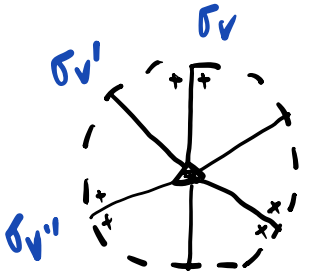
n个垂直镜面（两条实线）

n个群元

C_{nv} : $C_2, E, \sigma_v, \sigma_{v'}$

$\sigma_v = \sigma_{xz}(x, \bar{y}, z)$　（xz plane）

$\sigma_{v'} = \sigma_{yz}(\bar{x}, y, z)$　（yz plane）



C_{3v} : $C_3, C_3^2, E, \sigma_v, \sigma_{v'}, \sigma_{v''}$

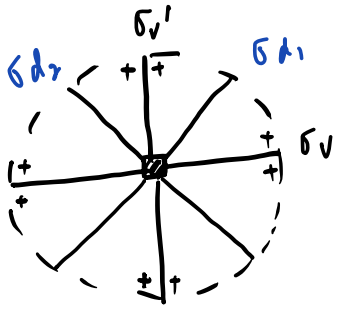
denote　$\sigma_v = \sigma_{xz}$

$\sigma_{v'} = C_{3z}\ \sigma_{xz}\left(-\frac{x}{2}+\frac{\sqrt{3}}{2}y,\ \frac{\sqrt{3}}{2}x+\frac{y}{2},\ z\right)$

$\sigma_{v''} = C_{3z}^{2}\ \sigma_{xz}\left(-\frac{x}{2}-\frac{\sqrt{3}}{2}y,\ -\frac{\sqrt{3}}{2}x+\frac{y}{2},\ z\right)$

$$\begin{pmatrix} \cos\frac{2\pi}{3} & -\sin\frac{2\pi}{3} & \\ \sin\frac{2\pi}{3} & \cos\frac{2\pi}{3} & \\ & & 1 \end{pmatrix}\begin{pmatrix} 1 & & \\ & -1 & \\ & & 1 \end{pmatrix}$$

C_{4v} :



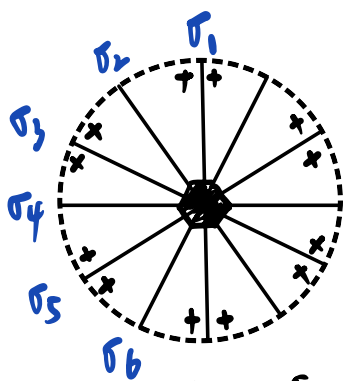
对角面
二面角面

C_4 , C_2, C_4^3 , E , σ_v, $\sigma_{v'}$, $\underline{\sigma_{d_1}}$, $\underline{\sigma_{d_2}}$

$\sigma_{d_1} = C_{4z}\ \sigma_{xz}\ (y, x, z)$ 由 $(\sigma_v, \sigma_{v'})$

$\sigma_{d_2} = C_{4z}^3\ \sigma_{xz}(\bar{y}, \bar{x}, z)$

C_{6v} :



C_6 C_3 , C_2, C_3^2, C_6^5 , E

$\sigma_v = \sigma_1 = \sigma_{xz}$ $\sigma_2 = C_{6z}\ \sigma_{xz}\ (\frac{x}{2} + \frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x - \frac{y}{2}, z)$

$\sigma_3 = C_{3z}\ \sigma_{xz} = \sigma_{v'}$ ($\dot{m}$ C_{3v})

$\sigma_4 = C_{2z}\ \sigma_{xz} = \sigma_{yz}$

$\sigma_5 = C_{3z}^2\ \sigma_{xz} = \sigma_{v''}$ ($\dot{m}$ C_{3v})

$\sigma_6 = C_{6z}^5\ \sigma_{xz}\ (\frac{x}{2} - \frac{\sqrt{3}}{2}y, -\frac{\sqrt{3}}{2}x - \frac{y}{2}, z)$

（国际符号，多用于晶体结构，更方便描述平移对称性）

Hermann - Mauguin 符号的基本特征是用不等价的轴或平面来标记晶体的对称性

轴：包含纯旋转轴与旋转反演轴

1, 2, 3, 4, 6　　$\bar{1}$ $\bar{2}$ $\bar{3}$ $\bar{4}$ $\bar{6}$

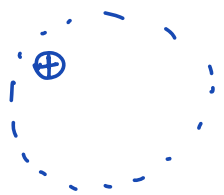
（更多用于分子对称性）

Schoenflies，旋转反演面，S_1, S_2, S_3, S_4, S_6

$S_1 = \bar{2}$，$S_2 = \bar{1}$，$S_3 = \bar{6}$，$S_4 = \bar{4}$，

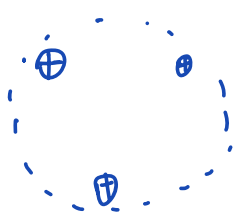
$S_6 = \bar{3}$　　$S_n = \sigma_h C_n$

$S_1 = \bar{2}$



$= C_{1h}$

$S_3 = \bar{6}$



$= C_{3h}$

$E, \bar{6}, (\bar{6})^2 = C_3$

$(\bar{6})^3 = \bar{2}$，$(\bar{6})^4 = C_3^2$

$(\bar{6})^5 = \bar{6}^5$

4. S_{2n}群 : （3个解）

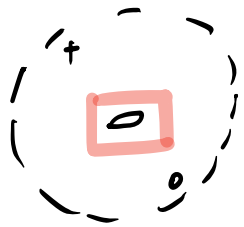
仅包含 n 度非正当旋转轴，且 n 为偶数

n 为奇数时与 C_{nh} 群相同. $S_1 = C_{1h}$

$S_3 = C_{3h}$

实.S：正当旋转

空.S：非正当旋转轴



S_2 : $S_2 = i$, E （C_i）

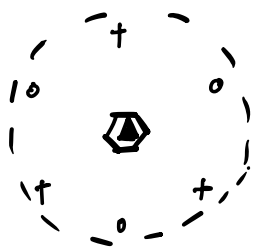
（parity）：空间反演



4 度非正当旋转

2 度正当旋转

S_4 : S_4, $S_4^2 = C_2$, $S_4^3 = S_4^{-1}$, E



6 度非正当旋转

3 度正当旋转

S_6 : S_6, $S_6^2 = C_3$, $S_6^3 = i$,

$S_6^4 = C_3^2$, $S_6^5 = S_6^{-1}$, $S_6^6 = E$

（比较 C_{nv} 见 P35）

5. D_n 群 : （4个）

包含一个 n 度主轴 和 n 个与之垂直的 2 度轴

$D_1 = C_2$（已出现）

（有1个2度轴）

D_2 :



C_{2z} , E

$C_{2x}(x, \bar{y}, \bar{z})$

$C_{2y}(\bar{x}, y, \bar{z})$

圆内 方解：与轴垂直的二度轴

实线 ：垂直纸面

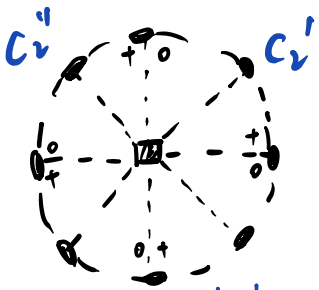


D_3 : C_{3z} , C_{3z}^2 , E, C_{2x}

$C_{3z}\ C_{2x}\left(-\frac{x}{2}+\frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x+\frac{y}{2}, \bar{z}\right) = C_2'$

$C_{3z}^2\ C_{2x}\left(-\frac{x}{2}-\frac{\sqrt{3}}{2}y, -\frac{\sqrt{3}}{2}x+\frac{y}{2}, \bar{z}\right) = C_2''$

圆上小椭圆：二度轴



4个二度轴，4为虚线

D_4 : C_{4z} , C_{2z} , C_{4z}^3 , E

C_{2x} , C_{2y} , $C_{4z}C_{2x} = C_2'(y, x, \bar{z})$

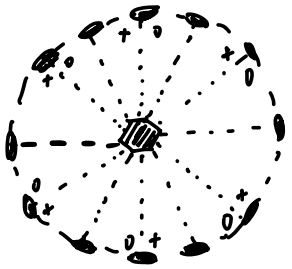
$C_{4z}^3\ C_{2x} = C_2''(\bar{y}, \bar{x}, \bar{z})$

$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

D_6 : C_{6z}, C_{3z} , C_{2z}, C_{3z}^2 , C_{6z}^5 , E, C_{2x} , C_{2y}

$C_{6z}\ C_{2x} = C_2^{①}$ (new)

$C_{3z}\ C_{2x} = C_2^{②} = C_2'$ (in D_3)



$C_{12}^{2}\ C_{2x} = C_2^{③} = C_{2}''$ (in D_3)

$C_{63}^{5}\ C_{2x}\left(\frac{x}{2} - \frac{\sqrt{3}}{2}y, -\frac{\sqrt{3}}{2}x - \frac{y}{2}, \bar{z}\right)$

$= C_2^{④}$ (new)

Summary: 正当旋动，水平镜面反射，垂直镜面反射，非正当旋动，与主轴垂直的二度轴（非常有规律的构造）

So far 21个群

b. D_{nh} 群

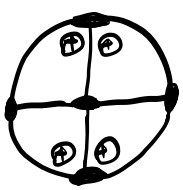
（4个群）

$D_n + \sigma_h$

$D_{1h} = C_{2v}$ (4 elements)

（一个不存在二度轴和包含二度轴的镜面 → C_{nv}

的主轴）



D_{2h} : C_{2z}, C_{2x}, C_{2y}, E

$C_{2z}\ \sigma_{xy} = i$, σ_{xy}

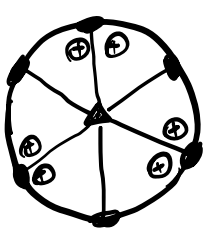
$C_{2x}\ \sigma_{xy}(x, \bar{y}, \bar{z}) = \sigma_{xz}$

$C_{2y}\ \sigma_{xy}(\bar{x}, y, \bar{z}) = \sigma_{yz}$

两线垂直成实线了

due to σ_{xz}, σ_{yz}

（联合效果相当于垂直镜面反射）



D_{3h} 全都在前面出现过

但不与某个群完全重合

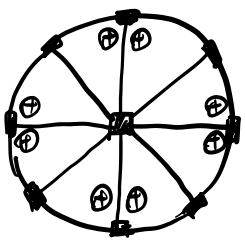
C_{3z}, C_{3z}^2, E, C_{2x}, $C_{3z}C_{2x} = C_2'$

$C_{3z}^2 C_{2x} = C_2''$, $C_{3z}\sigma_{xy} = S_{3z}$, $C_{3z}^2\sigma_{xy} = S_{3z}^{-1}$

σ_{xy}, $C_{2x}\sigma_{xy} = \sigma_{xz} = \sigma_v$,

$C_2'\sigma_{xy} = \sigma_v'$, $C_2''\sigma_{xy} = C_2''$

D_{4h}



可以比较简单的操作

O+ 生物体里，没出现过

才是新的

16个群元　8个 D_4 群的群元 ⊗ σ_h

D_4 群是 D_{4h} 的子群

C_4 群是 D_4 群的子群

C_{4z} C_{2z}, C_{4z}^3, E, C_{2x}, C_{2y}

$C_{4z}C_{2x} = C_2'$, $C_{4z}^3 C_{2x} = C_2''$

$C_{4z}\sigma_{xy} = S_{4z}$, $C_{2z}\sigma_{xy} = i$, $C_{4z}^3\sigma_{xy} = S_{4z}^{-1}$

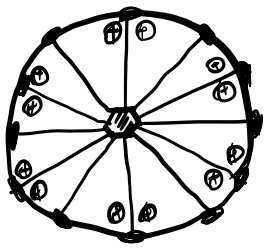
σ_{xy}, $C_{2x}\sigma_{xy} = \sigma_{xz}$, $C_{2y}\sigma_{xy} = \sigma_{yz}$

$C_2'\sigma_{xy}(y, x, z) = \underline{\sigma_{d_1}}$　$C_2''\sigma_{xy}(\bar{y}, \bar{x}, z) = \underline{\sigma_{d_2}}$

shown in C_{4v}

极射投影图

D_{6h}



24个群元

C_6 : C_{6z}, C_{3z}, C_{2z}, C_{3z}^2, C_{6z}^5, E

$C_{6z}C_{2x} = C_2^{①}$, $C_{3z}C_{1x} = C_2^{②} = C_2'$

$C_{3z}^2 C_{2x} = C_2^{③} = C_2''$, $C_{6z}^5 C_{2x} = C_2^{④}$

C_{2x}, C_{2y} (2度轴 ⊗ C_6)

$C_{6z}\sigma_{xy} = S_{6z}$, $C_{3z}\sigma_{xy} = S_{3z}$, $C_{2z}\sigma_{xy} = i$, $C_{3z}^2\sigma_{xy} = S_{3z}^{-1}$

$C_{6z}^5\sigma_{xy} = S_{6z}^{-1}$, σ_{xy}, (σ_h ⊗ C_6)

2度轴 ⊗ σ_h

$C_{2x}\sigma_{xy} = \sigma_{xz}$, $C_{2y}\sigma_{xy} = \sigma_{yz}$, $C_2'\sigma_{xy} = \sigma_v'$

$C_2''\sigma_{xy} = \sigma_v''$, $C_2^{①}\sigma_{xy}\left(\frac{x}{2}+\frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x-\frac{y}{2}, z\right) = \sigma_2$

$C_2^{④}\sigma_{xy}\left(\frac{x}{2}-\frac{\sqrt{3}}{2}y, -\frac{\sqrt{3}}{2}x-\frac{y}{2}, z\right) = \sigma_1$ (in C_{6v})

7. D_{nd} 群：

(n个群元)

D_{2d}, D_{3d}

D_n 群 与 垂直镜面 σ_d 组成 (n 度已当初动)

⇒ $2n$ 度非已当初动

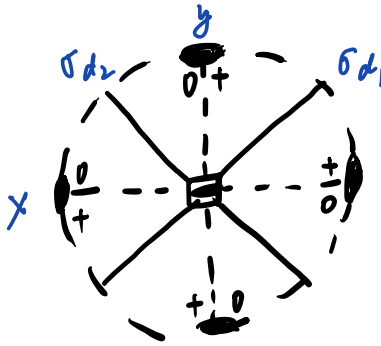
↓

$n = 1, 2, 3$

$n_{max} = 6$

$D_{1d} = C_{2v}$ (已出现)

没有主轴，那么一个2度轴作为主轴 vertical 镜面

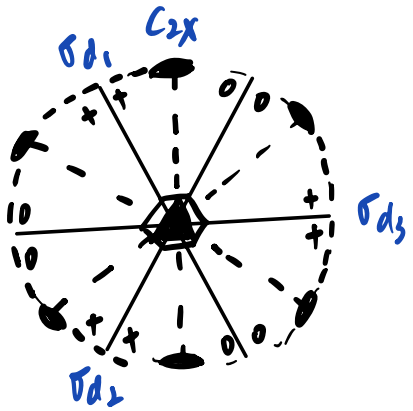


(4个群元)

D_{2d} : C_{2z}, C_{2x}, C_{xy}, E

σ_{d_1}, $C_{2z}\sigma_{d_1} = \sigma_{d_2}$

$C_{2x}\sigma_{d_1} = S_4^{-1}$, $C_{2y}\sigma_{d_1} = S_4$



$D_{3d} = D_3 \otimes \sigma_d$ (12个群元)

σ_d 是垂直镜面：

包含 C_3 轴

平分两个相邻 C_2 轴夹角的镜面

D_{3d} ： 无新操作

C_{3z}, C_{3z}^2, E, C_{2x}, $C_{3z}C_{2x} = C_2'$ ($\theta \to 60° - \theta$)

$C_{3z}^2 C_{2x} = C_2''$, $\sigma_{d_1}(\frac{x}{2}+\frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x-\frac{y}{2}, z)$ →

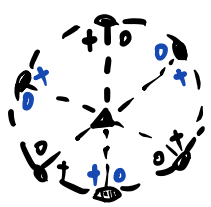
$C_{3z}\sigma_{d_1} = \sigma_{yz}$, $C_{3z}^2\sigma_{d_1}(\frac{x}{2}-\frac{\sqrt{3}}{2}y, -\frac{\sqrt{3}}{2}x-\frac{y}{2}, z) = \sigma_{d_2}$

$C_{2x}\sigma_{d_1}(\frac{x}{2}+\frac{\sqrt{3}}{2}y, -\frac{\sqrt{3}}{2}x+\frac{y}{2}, z)$ → ($\theta \to 240° - \theta$)

$C_2'\sigma_{d_1}(\frac{x}{2}-\frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x+\frac{y}{2}, z)$

$C_2''\sigma_{d_1} = i$

比较 $D_3 \otimes i$



这两个图是不一样的

二十七种点群

另外五个不能用极射投影图画

（没有主轴：至多两条）

↘ 最高度数的轴 （最高度数(n)）

和这轴有n个

8. 立方体群

T群：(23) 同构于

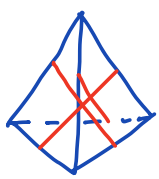
12个群元

(对称操作) (C_3)

E，3个二度轴，4个三度轴（体对角线）

4个三度轴负转动 (C_3^2)

四面体群，有24个



Th群：T群 + 空间反演

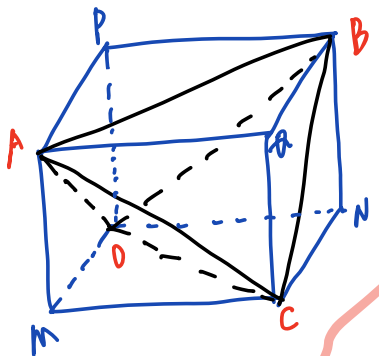
24个群元

Schoenflies

Td群：全部正四面体群，24个群元

OABC
(正四面体嵌入立方体)

单位元，8个关于体对角线的 C_3 操作
(顺时针)
3个关于 x, y, z 的 C_2
6个关于 x, y, z 的 S_4



$4C_3$, $4C_3^2$

σC_4, σC_4^{-1}
A→M→C

6个对角反射面

xy 面 $\perp \sigma_{d_1}$
$\perp \sigma_{d_2}$
同理可找 yz, xz 的对角反射面

xy, yz, xz

(ABMN) → σ_{d_1}, σ_{d_2} (OCP共)

立方体群 后2个

O 群：24个群元

(12条棱：对棱)

E, $8C_3$, $3C_2$, $6C_2$, $6C_4$

立方体 (四个体对角线) C_3, C_3^{-1}

(C_4 三对面 两心连线 C_4 & C_4^{-1})

过个体心中的 C_4^2 → $3C_2$

{T, T_h, T_d, O, O_h} 共5个

O_h 群：O群 + 空间反演

48个群元

C_n, C_{nh}, C_{nv}, S_{2m}, D_n, D_{nh}, D_{nd}

5 5 4 3 4 4 2